

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2015

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number** (not your name) on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
-

- Let G be a group of order p^2q , where p and q are distinct primes. Let H be a subgroup of G with $|H| = p^2$. Assume that H is not normal in G .
 - Show that H has q conjugates (including H).
 - Show that there is a homomorphism $\phi : G \rightarrow S_q$ (the permutations of q objects) such that pq divides $|\text{Im}\phi|$.
- Let V be a finite-dimensional vector space over a field F and let $T : V \rightarrow V$ be a linear transformation.
 - A cyclic vector for T is a vector $v \in V$ such that $\{v, Tv, T^2v, \dots\}$ spans V . Suppose that every nonzero vector $v \in V$ is a cyclic vector for T . Show that the characteristic polynomial of T must be irreducible over F .
 - Suppose that V is 2-dimensional over F and that the characteristic polynomial of T is irreducible over F . Show that every nonzero $v \in V$ is a cyclic vector for T .
- Let R be a commutative Noetherian ring with 1. Let $f = \sum_{n \geq 0} a_n X^n \in R[[X]]$ (=the ring of power series with coefficients in R). Show that there exists $N \geq 1$ such that $f^N = 0$ if and only if for each n there exists $N_n \geq 1$ such that $a_n^{N_n} = 0$. (*Hint:* For one implication, consider the ideal generated by $\{a_0, a_1, a_2, \dots\}$.)
- Let V and W be finite-dimensional vector spaces over $\mathbf{C}$. Let V^* be the dual space of V (=linear functionals $V \rightarrow \mathbf{C}$). Let $\text{Hom}(V^*, W)$ be the set of linear transformations from V^* to W . Consider the map
$$\phi : V \otimes W \rightarrow \text{Hom}(V^*, W)$$
defined by mapping $\sum v_i \otimes w_i$ to the linear transformation that sends $f \in V^*$ to $\sum f(v_i)w_i$. Show that ϕ is an isomorphism of vector spaces (you may assume that ϕ is a well-defined linear map). (You may not simply quote the result that such a map is an isomorphism, since this is what you are proving.)
- Let $f(X) \in \mathbf{Q}[X]$ be an irreducible polynomial of degree 4. Let r_1, r_2, r_3, r_4 be the roots of $f(X)$ in $\mathbf{C}$. Let

$$u = (r_1 + r_2)(r_3 + r_4), \quad v = (r_1 + r_3)(r_2 + r_4), \quad w = (r_1 + r_4)(r_2 + r_3)$$

and let

$$g(X) = (X - u)(X - v)(X - w).$$

- (a) Show that $u \neq v$ (similarly, $u \neq w$ and $v \neq w$, but you do not need to show these).
 (b) Show that $g(X) \in \mathbf{Q}[X]$.
 (c) Assume that $\text{Gal}(\mathbf{Q}(r_1, r_2, r_3, r_4)/\mathbf{Q}) \simeq S_4$. Show that $u, v, w \notin \mathbf{Q}$ and that $g(x)$ is irreducible.
 (d) Again assume that $\text{Gal}(\mathbf{Q}(r_1, r_2, r_3, r_4)/\mathbf{Q}) \simeq S_4$. Show that $\text{Gal}(\mathbf{Q}(u, v, w)/\mathbf{Q}) \simeq S_3$ (the permutations of 3 objects).

6. The following is the character table of a group G of order 12:

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	ω	ω^2
χ_3	1	1	ω^2	ω
χ_4	a	b	c	d

where ω is a primitive cube root of 1 and C_1, C_2, C_3, C_4 are conjugacy classes (with $C_1 = \{1\}$).

- (a) Find a, b, c, d .
 (b) Show that G has a normal subgroup of order 4.
 (c) Show that G does not have a normal subgroup of order 2.